

Two Failure Modes in Categorical Target-Leakage Screening: Sparsity Bias and Complete-Case Deletion

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ABSTRACT

Categorical target-leakage screening typically thresholds Cramér’s V . We show two independent failure modes on the same statistic, each traceable to a standard tooling default rather than to V itself. *Sparsity bias*: the Titanic name column scores $V = 0.999$ against survival despite carrying no information, because 1,305 of 1,307 names are singleton categories; a bias correction [2] collapses this to $\tilde{V} = 0.000$, while two well-behaved low-cardinality columns barely move (sex: $0.529 \rightarrow 0.528$; embarked: $0.184 \rightarrow 0.180$). *Complete-case deletion*: `pandas.crosstab`, and the PPS-based association used by `ydata-profiling` and `Deepchecks`’ `FeatureLabelCorrelation`, all drop rows with missing feature values before testing association—the default nearly everywhere. On Titanic’s boat column (a post-hoc leak, 62.9% missing, informative missingness), this drops the measured association from $V = 0.948$ (full data) to $V = 0.420$ (non-missing subset only), hiding the leak. We fix this in our own pipeline—encoding missingness as an explicit category, matching how mutual information already treats it—recovering $V = 0.951$ and raising our unified score from $\mathcal{L} = 0.739$ (warning) to $\mathcal{L} = 0.925$ (error). A negative replication on NSL-KDD (125,973 rows; three categorical features spanning cardinality 3–70, no missing values) confirms the sparsity mechanism requires genuine per-category sparsity, not cardinality alone: raw and corrected V are numerically indistinguishable there. Neither failure is fixed by tuning a threshold.

CCS CONCEPTS

• **Computing methodologies** → **Machine learning**; • **Mathematics of computing** → *Statistical paradigms*; • **Security and privacy** → *Software security engineering*.

KEYWORDS

target leakage, categorical data, Cramér’s V , bias correction, missing data, data quality

Key insight. Two standard defaults—chi-squared bias on sparse categories, and complete-case deletion of missing feature values—each break categorical leakage screening in a different, diagnosable way. One is fixed by a bias-corrected estimator; the other requires treating missingness itself as a feature. Neither fix addresses the other’s failure.

1 MOTIVATION

Standard categorical leakage screening thresholds an association statistic, usually Cramér’s V , between each feature and the target. Throughout this paper, *leakage* means *target leakage* [1]: information about the label that enters training but would not be available

at prediction time—not a security breach or data exfiltration. We show that on a single canonical dataset, this statistic fails in two directions for two distinct reasons, and that both reasons are properties of standard tooling defaults—not of V , and not idiosyncratic to our own implementation. The risk is not merely academic: near-unique identifier columns and administrative codes recorded post-outcome are common in exactly the security- and safety-relevant settings—fraud detection, credit scoring, clinical triage—where a missed leak is costly.

2 THE TWO FAILURES

Table 1 reports, for four Titanic columns (five rows, since boat appears both before and after the fix described below), cardinality, singleton-category count, raw and bias-corrected Cramér’s V , normalised mutual information $\tilde{I}$, performance inflation π , the unified score $\mathcal{L}$ computed from each version of the correlation term ($\mathcal{L}(V) = 0.35V + 0.35\tilde{I} + 0.30\pi$, and likewise for $\mathcal{L}(\tilde{V})$), and the ground-truth leakage label. Reporting $\mathcal{L}$ under both statistics matters: under raw V , name and sex are nearly indistinguishable despite one being leaky and the other clean; under corrected $\tilde{V}$ they cleanly separate. The two boat rows carry the same logic in reverse, discussed in Section 3.

Why name inflates. With 1,309 passengers and 1,307 distinct names, 1,305 names are singleton categories, and chi-squared-based association statistics are upward-biased whenever category counts are this sparse relative to sample size. The bias correction of Bergsma [2],

$$\tilde{V} = \sqrt{\frac{\max(0, \varphi^2 - \frac{(r-1)(c-1)}{n-1})}{\min(\tilde{r} - 1, \tilde{c} - 1)}}, \quad \tilde{r} = r - \frac{(r-1)^2}{n-1}, \quad \tilde{c} = c - \frac{(c-1)^2}{n-1},$$

collapses name to $\tilde{V} = 0.000$ —the estimator’s clipped floor (the $\max(0, \cdot)$ term), not a coincidentally exact zero—while sex and embarked (Table 1) barely move, confirming the correction targets sparsity specifically.

Why boat was missed, and by whom else. Lifeboat assignment was recorded only for passengers whose fate was already known, making boat a textbook post-hoc leak, yet on our original pipeline it scored only $V = 0.420$. Correlation is not blind to this leak in principle: 62.9% of boat is missing, and that missingness is itself almost deterministic of the outcome (Table 2)—98.1% of passengers with a boat number survived, versus 2.8% of those without one. `pandas.crosstab` excludes rows with a missing value by default: `ydata-profiling`’s own Cramér’s V implementation calls `pandas.crosstab` directly, and empirically, `Deepchecks`’ PPS-based `FeatureLabelCorrelation` scores boat at essentially zero (PPS = 0.0006) with real missing values present, versus 0.946 once missingness is an explicit indicator—the same failure, independently

Table 1: Five rows across four Titanic columns. name is a false positive under raw V ; boat is shown both before and after our missingness fix (Section 2), so the failure and the repair both appear here rather than in prose alone. boat’s cardinality differs between rows: pre-fix drops the 823 missing rows before counting categories (27); post-fix encodes missingness as its own category (28). [†]Clipped at the estimator’s floor, not an exact zero.

Feature	Card.	Sing.	V	$\tilde{V}$	$\tilde{I}$	π	$\mathcal{L}(V)$	$\mathcal{L}(\tilde{V})$	Truth
name	1307	1305	0.999	0.000 [†]	0.164	0.000	0.407	0.058	clean (FP under V)
boat (pre-fix)	27	4	0.420	0.351	1.000	0.808	0.739	0.715	leaky (FN under V)
boat (post-fix)	28	4	0.951	0.940	1.000	0.808	0.925	0.922	leaky (recovered)
sex	2	0	0.529	0.528	0.265	0.424	0.405	0.405	clean
embarked	3	0	0.190	0.184	0.054	0.000	0.085	0.083	clean

found. This is a shared default across the tooling ecosystem, not an isolated implementation bug.

Table 2: Where the original boat signal was lost. This table collapses boat to a binary missing/present indicator, isolating missingness’s own association; $V = 0.951$ elsewhere (Table 1) instead keeps all 27 non-missing categories distinct plus a 28th missing category, so the two are not directly comparable—both confirm the same signal survives, measured two ways.

Comparison	n	V
boat.notna() vs. survived (full data)	1309	0.948
boat vs. survived (non-missing only)	486	0.420

We fixed this in our own pipeline: missing feature values are now encoded as an explicit category rather than dropped, matching how mutual information already treats them (`pandas.factorize` assigns missing values their own code rather than NaN, so our MI and performance terms never dropped this signal—only correlation did). Table 1’s two boat rows show the result: $\mathcal{L}$ crosses from *warning* into *error* severity.

3 WHY MULTIPLE SIGNALS HELP—AND WHY ONE FIX DOES NOT COVER BOTH FAILURES

Before the fix, $\tilde{I}$ and π already saturated for boat while ρ did not (boat (pre-fix), Table 1: both $\mathcal{L}(V)$ and $\mathcal{L}(\tilde{V})$ clear 0.7 on $\tilde{I}$ and π alone), because mutual information estimation [3] never dropped the missing rows to begin with. This is a real, useful property of combining signals, but it is not the fix for boat: the two rows show the bias correction changes almost nothing for it, before the fix ($V=0.420 \rightarrow \tilde{V}=0.351$) or after ($V=0.951 \rightarrow \tilde{V}=0.940$)—sparsity correction and missingness-aware association are two different remedies for two different failures, and applying one does not substitute for the other. The weights (0.35, 0.35, 0.30) are a design choice, not a fitted parameter; we do not present the weighted sum itself as a research contribution.

4 NEGATIVE REPLICATION ON NSL-KDD

To test whether the sparsity mechanism generalises, we repeated the raw-vs.-corrected comparison on NSL-KDD [4] (125,973 rows), using its three native categorical features against a binary attack/normal label (46.5% attack rate). Table 3 reports the result: negative.

None of the three columns has any missing values, so this replication isolates the sparsity mechanism specifically. Even service,

Table 3: NSL-KDD: raw and corrected V are indistinguishable at every cardinality tested—no column here is sparse enough to trigger bias.

Feature	Card.	Sing.	V	$\tilde{V}$
protocol_type	3	0	0.282	0.282
service	70	1	0.860	0.860
flag	11	0	0.775	0.775

at 70 categories, has only 1 singleton among 125,973 rows—nothing like name’s 1,305 of 1,307—so the correction has nothing to remove. Despite service scoring as high as $V=0.860$, none of the three features is itself a leak by the same criterion that makes boat one: protocol_type, service, and flag are properties of a connection observable at classification time, not values recorded after the attack was labelled, so their association with the outcome is legitimately predictive rather than post-hoc. NSL-KDD therefore offers no false-positive or false-negative case here; it confirms instead that the sparsity mechanism tracks observations-per-category, not cardinality by itself.

5 PRACTICAL IMPLICATIONS

Any feature that is a discrete, deterministic function of the outcome—assignment codes, post-event labels, administrative flags—can evade a thresholded association statistic while remaining fully visible to mutual information and single-feature performance. We recommend three practices: (1) bias-corrected Cramér’s V on high-cardinality columns; (2) missingness indicators screened as features in their own right, since informative missingness survives complete-case deletion in every tool we tested; (3) never rely on a single statistic. Titanic remains one of the most widely used datasets in introductory ML instruction, and the screening practice it commonly illustrates is exactly the one shown insufficient here. The bias-corrected estimator and the full open-source pipeline (349 automated tests) are available at <https://github.com/jaimecano12/ml-framework>.

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